

Reproducible Regression Harness for Machine-Learning Security Tools

Contract-Validated Adapters Catching Cross-Repo Breakage Before It Ships

github.com/poojakira/mlsec-benchmark-suite (public, default branch main, primary language Python), MIT • Python 3.12 • HEAD 5c20f1da9602eb162efe8c63158da52b11d61f60 • verified 2026-10-01

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

1.00
HF-SCANNER F1

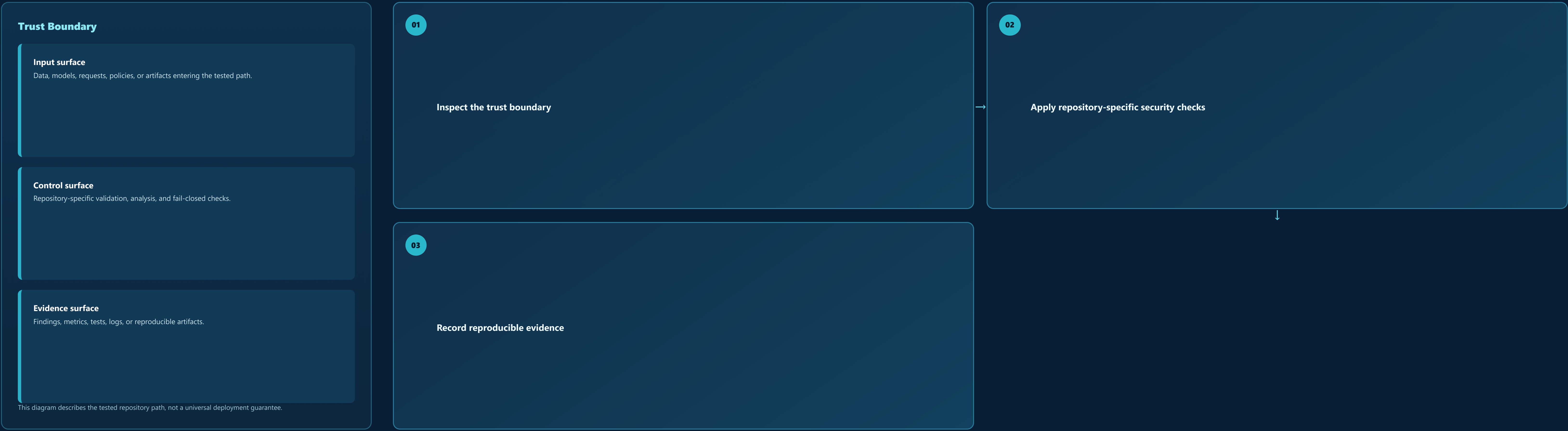
3/0
TP / FP

83 across 10 modules
Test functions

82 passed, 1 skipped
Current CI

SYSTEM VIEW

Architecture and Threat-Control Flow



Research Contribution

A contract-validated regression harness that wraps ML security tools in typed in-process adapters, runs them against versioned fixtures, and fails pytest on schema drift - with a hard, labeled separation between REAL product adapters and a synthetic smoke-test scaffold.

Verified Evidence

EV-01 VERIFIED_AT_SNAPSHOT - Real HF-scanner adapter: precision=recall=F1=1.0 on 5 fixtures - README verified real run tests/test_hf_scanner_adapter.py end-to-end (no mock); total_tp=3,total_fp=0,total_fn=0.	EV-02 VERIFIED_AT_SNAPSHOT - 83 test functions across 10 test modules; current Python 3.12 CI recorded 82 passed, 1 skipped with 90.07% statement coverage - repository README and test tree at HEAD 5c20f1da9602eb162efe8c63158da52b11d61f60.
EV-03 VERIFIED_AT_SNAPSHOT - JSON-schema contracts + versioned fixtures; pytest fails on drift - README how-it-works; schemas/ + contracts/ present.	EV-04 UNSUPPORTED (disclaimed by repo) - run-smoke per-category numbers - README: seeded-PRNG synthetic scaffold for plumbing, NOT measurements. Shown as synthetic only.
EV-05 UNSUPPORTED (disclaimed) - Real-world accuracy of wrapped tools - Suite validates contracts/format, not accuracy.	EV-06 Statement coverage: 90.07%

Method and Control Path

- 01 Inspect the trust boundary
- 02 Apply repository-specific security checks
- 03 Record reproducible evidence

Limitations and Residual Risk

- Validates contracts/format, not real-world accuracy.
- run-smoke numbers are synthetic (seeded PRNG).
- Fixture coverage is finite; gaps possible.
- Adapters can go stale if a tool changes internals.
- Real adapter gating requires sibling install.

Reproducibility

Evidence: results/real_hf_scanner.json, METHODOLOGY.md, tests/

Evidence Provenance

- Claim discipline**
Metrics come from the repository's verified metrics and research brief.
- Scope discipline**
Benchmarks are reported only at their documented fixture, dataset, or test scope.
- Reproduction**
PDF and PNG are generated locally from committed evidence without paid services.